

INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy $< 90\%$ $\rightarrow$ system fails
- If latency > 70 ms $\rightarrow$ unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy $< 95\%$ $\rightarrow$ defects may be missed
- If processing time > 50 ms $\rightarrow$ production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:

Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.

ANSWERS

1. Autonomous Vehicle Vision System Selection

Given Requirements:

The autonomous vehicle vision system has three strict constraints:

1. Accuracy must be at least 90%.
2. Latency must not exceed 70 ms.
3. The system supports only moderate computational resources.

The available methods are:

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Step 1: Evaluate Accuracy

The minimum required accuracy is 90%.

- Viola-Jones = 82% → Fail
- YOLO = 93% → Pass
- Deep Learning = 97% → Pass

Therefore, Viola-Jones can immediately be eliminated because its accuracy is below the required threshold.

Step 2: Evaluate Latency

The maximum acceptable latency is 70 ms.

- Viola-Jones = 30 ms → Pass
- YOLO = 60 ms → Pass
- Deep Learning = 90 ms → Fail

Although Deep Learning has the highest accuracy, its 90 ms latency is unacceptable for the autonomous vehicle.

Step 3: Evaluate Computational Requirement

The vehicle supports moderate computation.

- Viola-Jones $\rightarrow$ Low $\rightarrow$ Pass
- YOLO $\rightarrow$ Moderate $\rightarrow$ Pass
- Deep Learning $\rightarrow$ High $\rightarrow$ Fail

Deep Learning requires more computational resources than the vehicle can provide.

Overall Logical Evaluation

We can represent the selection condition as:

Accuracy $\geq 90\%$ AND Latency ≤ 70 ms AND Computation $\leq$ Moderate

Viola-Jones:

82% $\geq 90\%$ $\rightarrow$ False

Therefore:

False AND True AND True = False

✗ Rejected

YOLO:

93% $\geq 90\%$ $\rightarrow$ True

60 ms ≤ 70 ms $\rightarrow$ True

Moderate computation $\rightarrow$ True

Therefore:

True AND True AND True = True

☑ Selected

Deep Learning:

97% $\geq 90\%$ $\rightarrow$ True

90 ms ≤ 70 ms $\rightarrow$ False

High computation $\leq$ Moderate $\rightarrow$ False

Therefore:

True AND False AND False = False

✗ Rejected

Trade-Off Analysis

Viola-Jones provides excellent speed and low computational cost, but its accuracy is insufficient for an autonomous vehicle. In such a safety-sensitive application, missing an object could lead to incorrect vehicle decisions.

Deep Learning provides the highest accuracy and could detect objects more reliably. However, its 90 ms latency can delay decisions, and its high computational requirement is beyond the available hardware.

YOLO provides a balanced solution. Its 93% accuracy is above the required threshold, its 60 ms latency is within the limit, and its moderate computational requirement matches the available hardware.

Final Conclusion

YOLO should be selected for the autonomous vehicle vision system. It is the only method satisfying all three constraints simultaneously:

- Accuracy = 93% $\geq$ 90%
- Latency = 60 ms $\leq$ 70 ms
- Computation = Moderate

Therefore, YOLO offers the best combination of accuracy, real-time performance, and practical hardware requirements.

Final Choice: YOLO

2. Smart Factory Product Inspection Selection

Given Requirements:

The smart factory requires a computer vision system that can inspect products in real time.

The conditions are:

1. Accuracy must be at least 95%.
2. Processing time must not exceed 50 ms.
3. Hardware supports moderate computation.

Available methods:

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Step 1: Evaluate Accuracy

Required accuracy = 95%

- Traditional Image Analysis = 91% → Fail
- YOLO = 96% → Pass
- Deep Learning = 98% → Pass

Traditional Image Analysis cannot be selected because its accuracy is below the required value.

Step 2: Evaluate Processing Time

Maximum processing time = 50 ms

- Traditional Image Analysis = 25 ms → Pass
- YOLO = 45 ms → Pass
- Deep Learning = 75 ms → Fail

Deep Learning fails because its processing time is 25 ms higher than the permitted limit.

Step 3: Evaluate Computational Requirement

The available hardware supports moderate computation.

- Traditional Image Analysis → Low → Pass
- YOLO → Moderate → Pass
- Deep Learning → High → Fail

Therefore, Deep Learning is unsuitable for the given hardware.

Logical Evaluation

The selection condition is:

Accuracy $\geq 95\%$ AND Processing Time ≤ 50 ms AND Computation $\leq$ Moderate

Traditional Image Analysis:

Accuracy condition:

$91\% \geq 95\% = \text{False}$

Processing condition:

$25 \text{ ms} \leq 50 \text{ ms} = \text{True}$

Computation condition:

$\text{Low} \leq \text{Moderate} = \text{True}$

Overall:

$\text{False AND True AND True} = \text{False}$

✗ Rejected

YOLO:

Accuracy:

$96\% \geq 95\% = \text{True}$

Processing time:

$45 \text{ ms} \leq 50 \text{ ms} = \text{True}$

Computation:

$\text{Moderate} = \text{Moderate} = \text{True}$

Overall:

$\text{True AND True AND True} = \text{True}$

☑ Selected

Deep Learning:

Accuracy:

$98\% \geq 95\% = \text{True}$

Processing time:

$75 \text{ ms} \leq 50 \text{ ms} = \text{False}$

Computation:

$\text{High} \leq \text{Moderate} = \text{False}$

Overall:

True AND False AND False = False

✗ Rejected

Trade-Off Analysis

Traditional Image Analysis is very fast and inexpensive in terms of computation. However, its 91% accuracy is insufficient. In a manufacturing environment, this could result in defective products being classified as good products.

Deep Learning gives the highest accuracy of 98%, which is attractive for quality control. However, its 75 ms processing time can slow down the production line. Its high computational requirement also does not match the available hardware.

YOLO provides the best balance. It achieves 96% accuracy, which exceeds the required 95%, while its 45 ms processing time remains below the 50 ms limit. It also requires only moderate computation.

Final Conclusion

YOLO should be selected for the smart factory product inspection system.

It is the only method that satisfies all three requirements:

- Accuracy = 96% $\geq$ 95%
- Processing time = 45 ms $\leq$ 50 ms
- Computation = Moderate

Thus, YOLO provides an effective combination of high defect-detection accuracy, fast processing, and practical hardware requirements.

Final Choice: YOLO

Final Comparison:

Parameter	Autonomous Vehicle	Smart Factory
Minimum Accuracy	90%	95%
Maximum Latency/Time	70 ms	50 ms
Hardware	Moderate	Moderate
Selected Method	YOLO	YOLO
Accuracy	93%	96%
Latency/Time	60 ms	45 ms
Computation	Moderate	Moderate
Main Reason	Meets every constraint	Meets every constraint

Overall Answer

In both applications, YOLO is the optimal choice. Although Deep Learning provides higher accuracy, it fails the real-time and computational constraints. Viola-Jones and Traditional Image Analysis are faster and computationally cheaper, but their accuracy is below the required threshold. YOLO successfully satisfies all mandatory conditions, making it the most practical method for deployment.